E-Commerce Sales Dashboard

A Data Analytics Project Report

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Program:	B.E. Computer Science Engineering (Final Year)
Domain:	Data Analytics
Tools Used:	Python, MySQL, Power BI

1. Abstract

This project presents an end-to-end data analytics solution for an e-commerce business, covering the full pipeline from raw transactional data to executive decision-making. Using Python for data cleaning and feature engineering, MySQL for structured analytical querying, and Power BI for interactive visualization, the project analyzes over 51,000 simulated order records spanning four years (2022–2025) across six product categories and four regions of India. The analysis surfaces actionable insights on regional performance, seasonality, category profitability, customer value distribution, and the impact of discounting on profit margin, and translates these findings into concrete business recommendations.

2. Introduction

E-commerce businesses generate large volumes of transactional data daily, but raw data on its own rarely drives decisions — it must be cleaned, modeled, and visualized before it becomes useful to stakeholders. This project simulates that real-world workflow: starting from company-style raw exports (Orders, Customers, Products), the data is cleaned and merged into a single analysis-ready table, queried using SQL to answer specific business questions, and finally presented through an interactive Power BI dashboard designed for executive consumption.

3. Problem Statement

Business stakeholders need a fast, reliable way to answer questions such as: Which regions and product categories are most profitable? How does seasonality affect sales? Which customers drive the most value? And how do discounts affect margin? Raw transactional exports are too granular and inconsistent (duplicates, missing values, formatting issues) to answer these questions directly, creating the need for a structured cleaning, analysis, and visualization pipeline.

4. Objectives

- Simulate realistic, company-style raw e-commerce data (Orders, Customers, Products).
- Clean and engineer the data into a single analysis-ready dataset using Python.
- Perform structured analytical querying using MySQL, including window functions, views, and stored procedures.
- Design and build an interactive, multi-page Power BI dashboard with KPI cards, trend analysis, and drill-through.
- Derive and document business insights and recommendations from the analysis.

5. Methodology

5.1 Data Collection

Three raw datasets were generated to mimic a real company export: **Customers.csv** (customer demographics and location), **Products.csv** (product catalog with category, sub-category, and pricing), and **Orders.csv** (transaction-level records including sales, quantity, discount, profit, shipping cost, payment mode, and return status). The raw data was deliberately generated with realistic imperfections — duplicate rows, missing values, and a small number of invalid entries — to mirror the kind of cleanup a real analyst would need to perform.

5.2 Data Cleaning

The Python script **cleaning.py** loads all three raw files, removes duplicate records, imputes or drops missing values based on column-specific business rules, converts and validates date fields, standardizes text formatting, and removes records failing basic sanity checks (e.g., negative quantities, ship dates preceding order dates). The three tables are then merged into a single flat file, and four engineered columns are added: **Month**, **Year**, **Profit Margin**, **Sales Category**, and **Order Processing Days**. The final cleaned dataset contains approximately 51,300 records across 27 columns.

5.3 SQL Analysis

The cleaned dataset was loaded into a MySQL database (**ecommerce_dashboard**) and queried extensively. The query set covers core KPIs (total sales, profit, average order value), time-based analysis (monthly/yearly sales, year-over-year growth using window functions), category and sub-category breakdowns, top-N rankings using **RANK()**, **DENSE_RANK()**, and **ROW_NUMBER()**, customer segmentation, regional comparisons, reusable **views**, and parameterized **stored procedures** for repeatable analysis.

5.4 Power BI Dashboard

A five-page executive dashboard was designed: Executive Summary, Sales Analysis, Customer Analysis, Product Analysis, and Regional Analysis. The data model uses a dedicated date dimension table to support time-intelligence DAX measures (YTD, YoY growth, rolling averages). Interactivity features include synchronized slicers, drill-through from category to product level, bookmarks for guided views, and custom tooltips. Full build instructions and the complete DAX measure library are documented separately in **PowerBI_Build_Guide.md**.

6. Business Insights

Analysis of the cleaned dataset (~■301.5 Cr total sales, ~■62.8 Cr total profit, 20.8% overall margin) revealed the following:

- **Regional performance:** the South region leads in total sales, while the East region trails roughly 42% behind, indicating an opportunity for targeted market expansion.
- **Seasonality:** August is the strongest sales month and February the weakest, suggesting promotional campaigns should target the February dip and inventory should scale ahead of the July–August peak.

- **Category profitability:** Electronics is the most profitable category by a wide margin, while Office Supplies contributes volume but minimal profit.
- **Discount impact:** average profit margin falls from ~29.8% on full-price orders to ~6.5% on orders discounted above 15%, showing a strong inverse relationship between discount depth and profitability.
- **Customer concentration:** a small set of top customers each contribute well over ■1 crore in lifetime sales, highlighting the value of a tiered loyalty strategy.

7. Conclusion

This project demonstrates a complete, industry-style analytics workflow — from raw data generation through cleaning, structured SQL analysis, and interactive dashboarding — and shows how each stage builds toward answering concrete business questions. The resulting dashboard and insights provide a template that could be adapted to a real e-commerce business's own transactional data with minimal modification to the pipeline.

8. References

- Pandas Documentation – https://pandas.pydata.org/docs/
- MySQL 8.0 Reference Manual – https://dev.mysql.com/doc/refman/8.0/en/
- Microsoft Power BI Documentation – https://learn.microsoft.com/power-bi/
- DAX Function Reference – https://learn.microsoft.com/dax/